

Coupled socio-ecological modelling of land-use dynamics and erosion risks in Bolango Watershed, Indonesia

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Abstract: This study examines how land-use change alters soil erosion in the Bolango Watershed, Gorontalo Province, Indonesia, through an integrated framework that couples the revised universal soil loss equation (RUSLE) with agent-based modelling (ABM). Across the 52,494 ha watershed, marked transformations were projected between 2013 and 2033, including a 14.25% decline in forest cover, a 42.37% expansion of dryland agriculture, and a 122.24% increase in settlements. Spatial analyses implemented in geographic information system (GIS) software showed that mean soil loss decreased from 59.08 Mg·ha⁻¹·y⁻¹ in 2013 to 49.22 Mg·ha⁻¹·y⁻¹ in 2023, before rising slightly to 50.17 Mg·ha⁻¹·y⁻¹ in 2033. Rainfall erosivity (*R*) declined from 921.56 to 760.80 MJ·mm·ha⁻¹·h⁻¹·y⁻¹, whereas the slope length–steepness factor (*LS*) ranged from 0.1 to 276.82, underscoring strong terrain control. Scenario simulations showed that contour farming reduced total erosion by 27%, while strict protection of slopes >25% reduced erosion by 51%, demonstrating the effectiveness of targeted conservation measures. The ABM represented interactions among farmers, government, and investors and showed that policy enforcement and environmental awareness strongly shape land-use decisions. Model validation yielded $R^2 = 0.83$ and $RMSE = 12.7$ Mg·ha⁻¹·y⁻¹, indicating robust predictive performance. The coupled ABM–RUSLE framework therefore provides a rigorous basis for sustainable watershed management and spatial planning in tropical environments.

Keywords: agent-based modelling, Bolango Watershed, geographic information system (GIS), land-use change, patch-generating land use simulation (PLUS), soil erosion

INTRODUCTION

Land-use and land-cover change (LULCC) is a major threat to watershed sustainability, particularly in tropical and semi-humid regions. Converting natural landscapes to agricultural and urban land disrupts hydrological pathways, increases surface runoff, and accelerates soil erosion. In tropical watersheds, these effects are intensified by frequent high-intensity rainfall, which mobilises sediment, degrades water quality, and weakens ecosystem services.

Recent studies show that LULCC alters sediment yield, infiltration, and streamflow dynamics, with direct consequences for ecological stability and the socio-economic well-being of local communities (Getu Engida *et al.*, 2021; Negese, 2021; Payuyu, 2022; Payuyu, 2023; Ayalew, Niguse and Aragaw, 2024). In Gorontalo Province, Indonesia, agricultural expansion has intensified erosion, increased flood and landslide risk, and reduced river-basin carrying capacity.

The Bolango Watershed in Gorontalo Province, covering approximately 52,471.44 ha, exemplifies these linked environ-

mental pressures. Its fan-shaped geomorphology, with a wide upstream contributing area and a constricted downstream section, promotes runoff concentration and heightens flood risk downstream (Habir, 2022). Recent evidence from the Bolango River further shows that total suspended solids (TSS) exceed the thresholds set by Peraturan (2021), indicating active erosion and sediment delivery within the basin (Mosi *et al.*, 2024). Concurrent land clearing, expansion of dryland agriculture, and unregulated residential growth have collectively reduced watershed resilience. These conditions make Bolango an appropriate case for an integrated, dynamic modelling framework capable of diagnosing present erosion patterns and projecting future responses to land-use change.

Spatial planning in Gorontalo Province, particularly through the regional spatial plan, is an important institutional driver of land-cover change because it determines protection and cultivation zones, guides infrastructure and settlement corridors, and conditions permits for agricultural conversion and urban development. In the Bolango Watershed, these planning directives interact with economic incentives and population pressure; consequently, differences in compliance and enforcement can produce measurable shifts from forest to agriculture and settlement, especially along accessible valley corridors and on lower-to-mid slopes.

Previous studies have widely applied the universal soil loss equation (USLE) and the revised universal soil loss equation (RUSLE) to estimate soil loss from rainfall erosivity (R), soil erodibility (K), topographic controls (LS), cover-management effects (C), and conservation practices (P). Integrated with geographic information systems (GIS), these models have been effective for mapping erosion risk and identifying critical land areas (Arsyad, 2010; Benavidez *et al.*, 2018; Lihawa *et al.*, 2025). Their main limitation, however, is that conventional GIS–RUSLE applications are essentially static: they describe erosion under fixed conditions and cannot represent dynamic socio-ecological interactions. Because land use changes in response to human decisions, economic pressure, and policy intervention, static models cannot capture feedbacks between behaviour and biophysical change and therefore remain limited for long-term forecasting (Borrelli *et al.*, 2020; Weng *et al.*, 2023).

These limitations have encouraged a shift toward integrating process-based and behaviour-based approaches. Agent-based modelling (ABM) provides a powerful framework for representing the decisions of multiple actors – such as farmers, local governments, and investors – whose cumulative actions reshape landscapes over time (Majumdar, Shukla and Maiti, 2018). In environmental applications, ABM can capture adaptive behaviour, policy responses, and cross-scale spatial interactions, thereby complementing empirical erosion models. Recent studies (Hazbavi, Sadeghi and Gholamalifard, 2019; Borrelli *et al.*, 2020; Keller *et al.*, 2021) show that coupling ABM with RUSLE can generate more realistic simulations of land-use dynamics and can improve the evaluation of alternative management scenarios. This hybrid perspective is especially valuable for analysing long-term feedbacks between socioeconomic drivers and physical erosion processes.

In this study, the ABM operationalises qualitative behavioural concepts – including environmental awareness, economic motivation, and policy compliance – by translating them into explicit decision rules and transition probabilities for land-use conversion and conservation adoption. The quantitative link to

RUSLE is established by updating land-cover-dependent erosion parameters, primarily the C and P factors, at each simulation step so that agent decisions directly modify the biophysical inputs used to calculate soil loss over time.

Within this framework, hydro-morphometric parameters are central to watershed behaviour and carrying capacity (Kamaraj *et al.*, 2024). Topography, soil texture, rainfall intensity, and vegetation cover interact to determine erosion susceptibility, while socioeconomic drivers such as agricultural expansion, population pressure, and policy enforcement act as indirect but powerful controls on land degradation (Lihawa and Sutkino, 2009; Abu Hammad and Tumeizi, 2012; Salote Lihawa and Dunggio, 2022; Machmud *et al.*, 2023). Together, these findings indicate that sustainable watershed management in rapidly developing regions must be grounded in both biophysical conditions and human decision-making.

Although GIS–RUSLE remains useful for diagnostic erosion mapping, the absence of a dynamic decision component limits its value for long-term watershed planning. This limitation has motivated the adoption of ABM–RUSLE integration, which allows for iterative simulation of actor decisions and biophysical responses over time. The ABM framework offers the flexibility to represent stakeholder objectives, perceptions, and adaptive strategies under alternative land-use policies. By coupling those dynamics with RUSLE, the model can repeatedly update the C and P factors as land-use states change, thereby increasing predictive realism. This is particularly relevant in the Bolango Watershed, where decisions made by smallholder farmers and urban developers directly influence the spatial pattern and intensity of erosion.

This study develops a spatially explicit ABM–RUSLE model to simulate reciprocal interactions between land-use dynamics and erosion rates in the Bolango Watershed. The model combines empirical erosion estimation with agent-based behavioural rules to represent stakeholders, especially farmers and policymakers, whose decisions influence landscape configuration over time. Using a 10–30-year simulation horizon, the framework projects cumulative soil loss, expansion of degraded land, and the likely effectiveness of conservation policies. By linking micro-level decisions to macro-level environmental outcomes through iterative feedbacks, the model addresses a key gap in current watershed research.

The model is spatially explicit at two complementary scales. At the micro scale, landscape states and agent decisions are represented as grid cells (patches/land parcels) that store land-cover classes together with associated C and P values. At the macro scale, outputs are summarised for the Mongiilo, Bolango Hilir, Langge, and Dulamayo sub-watersheds and for the watershed as a whole, allowing headwater processes to be linked to downstream implications within a micro-watershed perspective.

The integration is grounded in established theoretical and methodological work. The RUSLE model remains the standard framework for soil-loss estimation (Renard *et al.*, 1991), with validated derivations for rainfall erosivity (Babu, Dhyani and Kumar, 2004; Jain and Das, 2010), soil erodibility (Bouguerra *et al.*, 2017), and topographic controls (David, 1988). Empirical coefficients for land cover and conservation practices were adapted from regional studies (Asdak, 1995; Arsyad, 2010; Benavidez *et al.*, 2018). At the same time, ABM platforms such as NetLogo have expanded the capacity to simulate spatially

heterogeneous decision-making. Bringing these components together in a unified framework represents a meaningful advance in watershed simulation science.

The Bolango Watershed provides a compelling test case that demonstrates why innovative coupled modelling is needed to prevent renewed erosion escalation, even where partial remediation efforts are already underway. Empirical evidence from the watershed underscores the urgency of such an approach. Rainfall erosivity ranged from 613.18 to 921.56 MJ·mm·ha⁻¹·h⁻¹·y⁻¹ in 2013 and forecast values for the 2023–2033 period range from 582.63 to 760.80. The soil erodibility factor (*K*) varied from 0.0264 to 0.0394 according to soil texture, while the *LS* factor ranged from 0.1 to 276.82, emphasising the role of steep slopes in amplifying erosion potential. Land-cover-dependent *C* and *P* factors also varied substantially among agricultural, forested, and residential areas. The patch-generating land use simulation (PLUS) model projected land-use transitions from 2013 to 2033, including a 14% decline in forest area, a 42% increase in dryland agriculture, and a 122% expansion of residential land. Correspondingly, average erosion decreased from 59.08 Mg·ha⁻¹·y⁻¹ in 2013 to 49.22 Mg·ha⁻¹·y⁻¹ in 2023, but rose slightly to 50.17 Mg·ha⁻¹·y⁻¹ in 2033, indicating that degradation pressure persists despite partial improvements.

These patterns indicate that forest loss, especially where it is followed by agricultural intensification, is a primary driver of erosion in the Bolango Watershed. They also show that the interaction between land-use change and hydrological response is not linear, underscoring the difficulty of achieving sustainable watershed management under rapid development. Integrating ABM with RUSLE therefore does more than improve prediction; it strengthens decision support by enabling iterative evaluation of interventions such as reforestation, contour farming, and steep-slope protection under alternative policy settings.

The novelty of this research lies in coupling agent-based behavioural dynamics with an established empirical erosion model to create an interactive, spatially explicit decision-support system. This integration bridges environmental modelling and social simulation, allowing the bidirectional feedback between land-use decisions and erosion processes to be quantified more explicitly. As a result, the model offers a more comprehensive basis for evaluating alternative land-management pathways and for designing interventions that balance ecological integrity with socio-economic development.

This research therefore aims to: (1) model the spatial and temporal interactions between land-use change and soil erosion using an integrated ABM–RUSLE framework; (2) simulate alternative management scenarios to assess the long-term sustainability of the Bolango Watershed; and (3) generate spatially explicit recommendations for conservation policy and land-use planning. Beyond its methodological contribution, the study is intended to support regional watershed management strategies that explicitly integrate human and biophysical systems in pursuit of sustainable development.

MATERIALS AND METHODS

This study applies a quantitative, spatially explicit modelling framework that couples the revised universal soil loss equation (RUSLE) with agent-based modelling (ABM) to simulate dynamic

interactions between land-use change and soil erosion in the Bolango Watershed, Gorontalo Province. The design combines empirical erosion estimation, land-use projection, and behavioural simulation to evaluate future erosion trajectories under alternative management strategies.

The study area covers the Bolango Watershed, which spans Bone Bolango Regency, Gorontalo Regency, and Gorontalo City. Geographically, the watershed extends from 0°29'42.59" to 0°48'23.74" N and from 123°00'51.91" to 123°15'56.61" E, with a total area of approximately 52,471.44 ha (Fig. 1).

The watershed boundary, administrative setting, and sub-watershed units are used for analysis and reporting. Spatial and non-spatial datasets were assembled from authoritative national and international sources. Watershed boundary maps were obtained from the Ministry of Environment and Forestry at a scale of 1:50,000. Rainfall data were derived from the Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS) product of the University of California, Santa Barbara Climate Hazards Group at 5,566 m resolution. Soil information was obtained from EnvirometriX Ltd. at 250 m resolution, while topographic data were derived from NASA JPL Shuttle Radar Topography Mission (SRTM) data at 30 m resolution. Land-cover maps for 2013 and 2023 were acquired from the Ministry of Environment and Forestry at a scale of 1:250,000. All datasets were harmonised in a geographic information system (GIS) environment and processed using QGIS software.

To improve methodological transparency and reproducibility, all datasets were harmonised to a common spatial framework before factor derivation. Terrain-dependent variables, especially slope and flow accumulation, were derived from the 30 m digital elevation model (DEM) to preserve the detail needed to estimate the topographic factor representing slope length and steepness (*LS*), whereas coarser layers such as soil grids were resampled to the analysis grid using consistent GIS procedures. Parameter values for *R*, *K*, *LS*, *C*, and *P* are reported as spatial ranges and linked to the corresponding data sources and equations so that input derivation and error propagation are explicit.

The revised universal soil loss equation (RUSLE) developed by Renard *et al.* (1991) was used to estimate soil erosion rates using the empirical equation:

$$A = R \cdot K \cdot LS \cdot C \cdot P \quad (1)$$

where: *A* = the annual average soil loss (Mg·ha⁻¹·y⁻¹), *R* = rainfall erosivity (MJ·mm·ha⁻¹·h⁻¹·y⁻¹), *K* = soil erodibility (Mg·ha·h·ha⁻¹·MJ⁻¹·mm⁻¹), *LS* = the topographic factor representing slope length and steepness (–), *C* = the cover-management factor (–), *P* = the conservation-practice factor (–). Each component was derived as follows.

Rainfall erosivity was calculated using the equation of Babu, Dhyani and Kumar (2004), as modified by Jain and Das (2010), for annual precipitation totals between 340 and 3,500 mm. The CHIRPS data for 2013–2023 were processed to derive *R* values, which ranged from 613.18 to 921.56 in 2013 and from 582.63 to 760.80 MJ·mm·ha⁻¹·h⁻¹·y⁻¹ in 2023–2033. This formulation is widely used in regional erosion studies because it represents spatial rainfall variability reliably (Pindozzi *et al.*, 2017).

The *K* factor was derived from soil-texture and organic-content information following Bouguerra *et al.* (2017). Values of

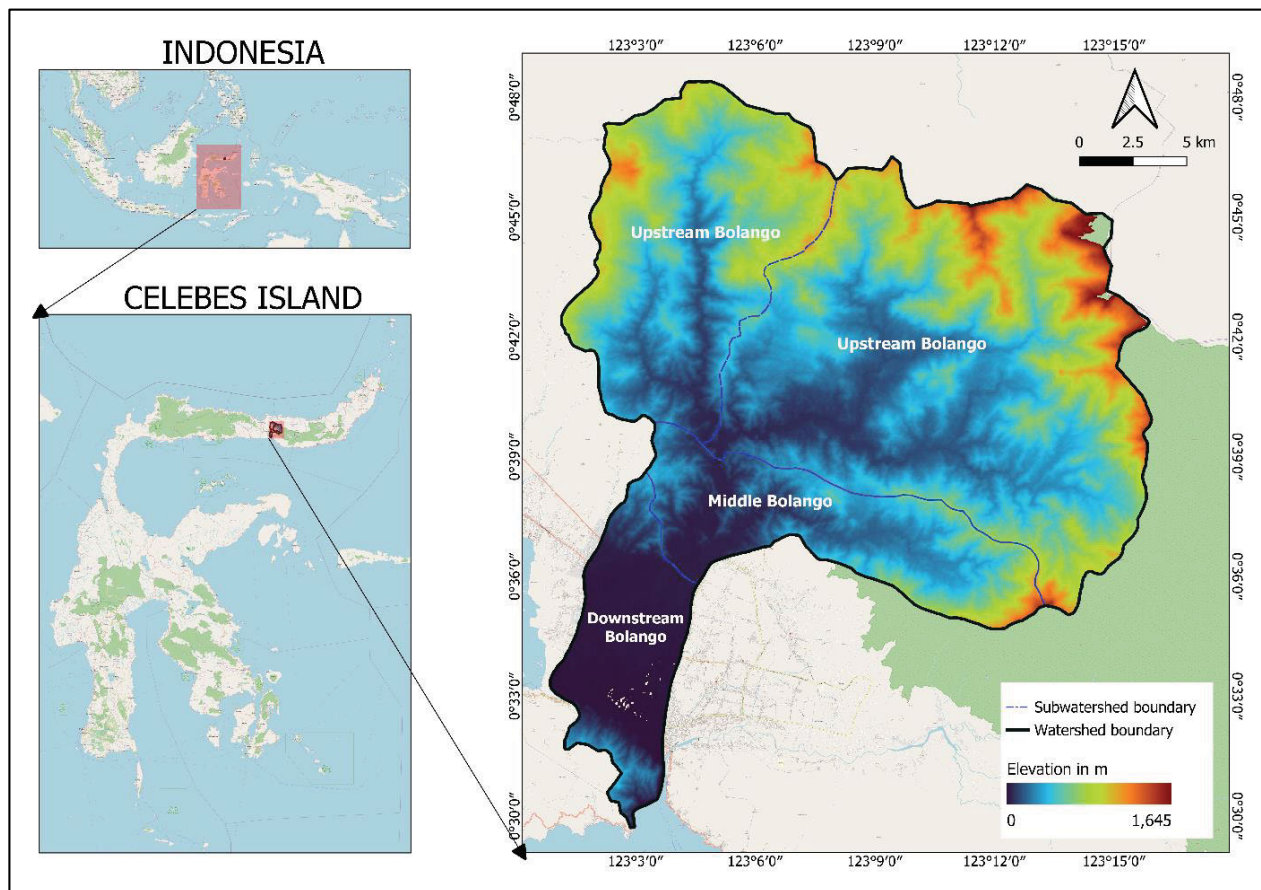


Fig. 1. Location of the Bolango Watershed in Gorontalo Province, Indonesia; source: own elaboration

K were extracted from OpenLandMap and linked to United States Department of Agriculture (USDA) soil-texture classes. Representative values included 0.0288 for clay, 0.0360 for sandy clay, 0.0394 for clay loam, and 0.0264 for sandy clay loam. Comparable data-driven soil mapping approaches have been shown to improve the spatial accuracy of erosion assessments in tropical environments (Hua, 2017).

The LS factor was calculated from the 30 m SRTM DEM using the formulation proposed by David (1988). Resulting values ranged from 0.1 to 276.82, reflecting the strong relief contrast of the watershed. Terrain analysis was implemented in QGIS, where slope and flow accumulation were derived using hydrological processing algorithms (Xu *et al.*, 2023).

This wide LS range occurs within a single watershed because the terrain varies sharply from steep headwater hillslopes to relatively flat downstream plains. The highest LS values are concentrated in upper-catchment ridge-to-channel slopes, particularly in parts of the Mongiilo and Dulamayo sub-watersheds, whereas low values dominate valley bottoms and lower-gradient terrain. This contrast explains the pronounced watershed-wide heterogeneity of topographic erosion potential.

The C factors were assigned to land-use classes using empirical coefficients from Asdak (1995) and Arsyad (2010). Satellite-derived normalised difference vegetation index ($NDVI$) was used to refine the C factor estimates and to represent temporal variation in vegetation cover more dynamically. Typical C values ranged from 0.001 in primary forest to 0.9 in built-up land, which is consistent with broader RUSLE applications (Li *et al.*, 2022).

The P factors were assigned according to land-use type and slope class following Benavidez *et al.* (2018) and David (1988). The resulting P values ranged from 0.005 in protected forest to 0.95 in urbanised or open land. Contour farming, terracing, and slope protection were incorporated into the scenarios as alternative P values, enabling conservation practices to be represented explicitly in the simulations. This structure reflects the real-world role of management interventions in stabilising soil and reducing erosion (Benavidez *et al.*, 2018).

Because the P factor is assigned categorically by land-cover type and slope class, a single watershed-average value would mask management-relevant heterogeneity. We therefore report the full baseline range and document the slope-class assignments used in scenario 2 (contour farming) and scenario 3 (protected steep slopes), enabling reproduction of P summaries for any sub-area of interest.

The patch-generating land use simulation (PLUS) model was used to project land-use distribution for 2033. The model integrates the land expansion analysis strategy (LEAS) and cellular automata based on random seeds (CARS) to simulate patch-level landscape change (Liang *et al.*, 2021). Predictor variables included land cover in 2013 and 2023, slope, elevation, distance to roads, vegetation index, and night-time lights. Model performance was evaluated using the kappa coefficient and figure of merit (FOM), which reached 0.89 and 0.68, respectively, matching validation ranges reported in comparable applications (Zhang *et al.*, 2024; Liu *et al.*, 2025). The 2033 land-use projection was then used as the basis for scenario-based erosion modelling.

Three scenarios were developed to evaluate erosion under alternative management strategies. Scenario 1 (business-as-usual)

used the 2033 projected land cover without additional conservation measures. Scenario 2 represented contour-farming adoption and assigned P values from 0.6 on 9–12° slopes to 0.95 on slopes >25°, following David (1988). Scenario 3 imposed strict protection on slopes >25%, designating those areas as protected zones with $P = 0.005$ in accordance with Indonesian law (Undang-Undang, 2007). Together, the three scenarios allowed the relative effectiveness of conservation interventions to be compared under consistent biophysical conditions.

The 2033 simulations are explicitly scenario-based. Over the 10-year projection horizon, topography (LS) and soil properties (K) were treated as time-invariant, rainfall erosivity (R) followed the CHIRPS-derived pattern used for 2023–2033, and the land-cover-dependent C and P factors were allowed to change in response to simulated land-use transitions and management rules. The scenario analysis therefore isolates how alternative policy and practice settings – business-as-usual, contour farming, and steep-slope protection – modify erosion outcomes under otherwise comparable environmental conditions.

The ABM was used to simulate the decisions of actors who influence land use and erosion dynamics. Agents were represented primarily as farmers and government officials, each characterised by attributes such as environmental awareness, economic motivation, social norms, technical capacity, and policy compliance (Souza Silva da *et al.*, 2021; Velmurugan, Adamala and Palanivel, 2021). The model includes a feedback loop in which land-use change modifies the C and P factors, alters annual average soil loss (A), and subsequently influences agents' perceptions and future decisions. Implemented in a cellular automata framework, the model simulates a 10–30-year period through iterative adjustments of land use and erosion parameters. Similar frameworks have been successfully applied to socio-environmental systems to evaluate adaptive conservation behaviour (Nampak *et al.*, 2018).

The analytical workflow is summarised in Figure 2. First, environmental datasets were processed to map the RUSLE factors (R , K , LS , C , and P) across the watershed. The PLUS model was then used to simulate land-use change through 2033. Next, the land-cover-dependent RUSLE factors were updated under the projected conditions, and erosion rates were estimated for each management scenario. The coupled ABM–RUSLE model was then used to generate iterative simulations in which human decisions and erosion outcomes influenced one another through feedback.

Model performance was evaluated using the kappa coefficient, coefficient of determination (R^2), and root mean square error ($RMSE$) (Koirala *et al.*, 2019; Thapa, 2020; Kwang *et al.*, 2022). Spatial validation compared predicted erosion patterns with historical observations to assess model consistency and accuracy.

The institutional pathway embedded in the model is also clarified in Figure 2. Farmer agents implement on-farm practices, such as contour planting, through local farmer groups and extension support, whereas government agents operationalise spatial planning through zoning, permit control, and compliance monitoring. These interactions are encoded as scenario-specific rules – business-as-usual, conservation incentives, or strict protection – so that the model can evaluate how policy choices cascade into land-cover change and, ultimately, erosion outcomes.

This integrated methodology provides a robust framework for understanding how socio-economic drivers and physical processes interact in the Bolango Watershed. By coupling empirical erosion modelling, land-use projection, and behavioural simulation, the study delivers a dynamic assessment of future watershed conditions under alternative management settings and supports evidence-based land-use planning.

Uncertainty and model limitations were considered explicitly when interpreting the results. RUSLE is an empirical sheet-and-rill erosion model and does not represent gully erosion, mass wasting, or channel sediment transport; consequently, absolute soil-loss values may be conservative or incomplete in steep terrain. In addition, factor derivation – especially for C and P – depends on land-cover classification and generalised coefficients, while the 2033 projections inherit uncertainty from both the PLUS simulation and the simplified behavioural rules in the ABM. Accordingly, the scenario results are interpreted primarily as relative differences among management options rather than exact year-specific forecasts.

RESULTS AND DISCUSSION

LAND USE CHANGE (2013–2033)

Land-use dynamics in the Bolango Watershed between 2013 and 2033 reveal substantial redistribution among the main land-cover classes. As shown in Figure 3, forest area declined from 26,781 ha in 2013 to 22,963 ha in 2033, equivalent to a net loss of 14.25%.

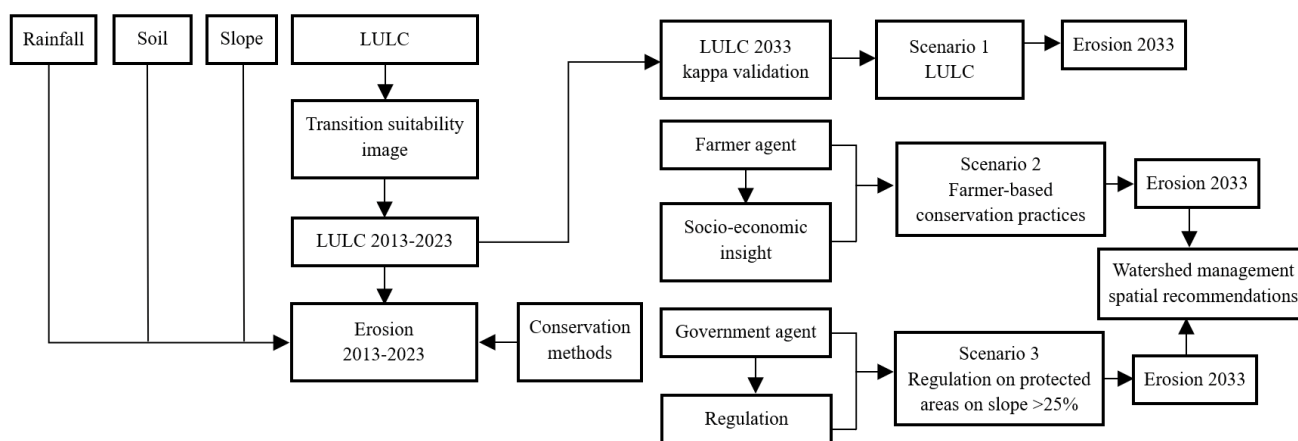


Fig. 2. Conceptual workflow of the coupled agent-based modelling – revised universal soil loss equation (ABM–RUSLE) framework; LULC = land use / land cover; scenarios 1–3 as in pp. 235–236; source: own elaboration

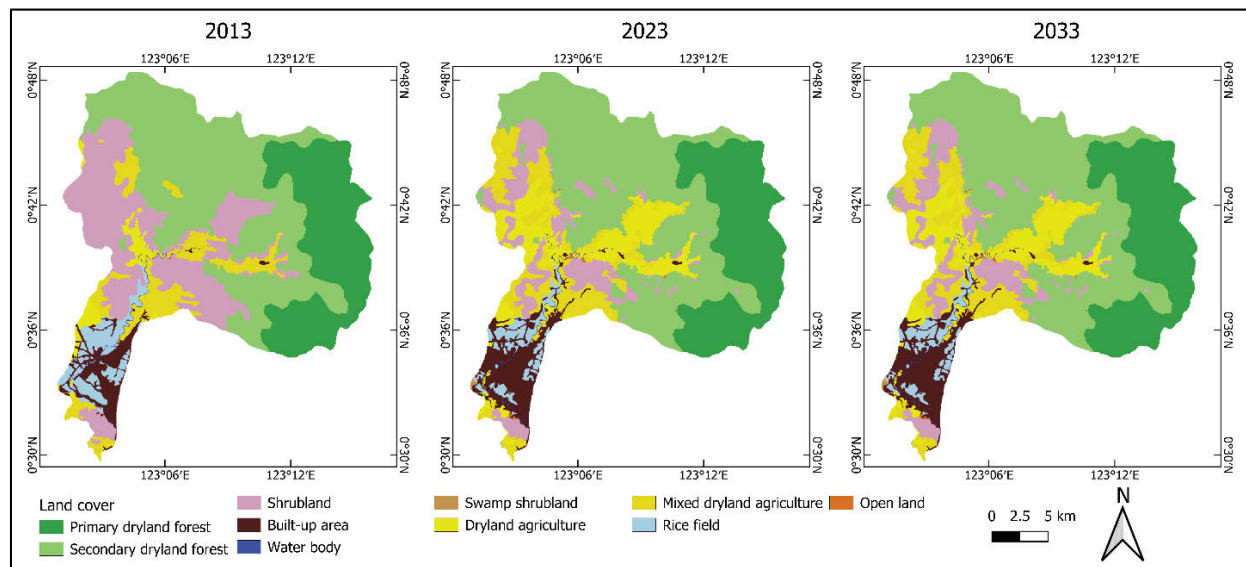


Fig. 3. Observed land-use/land-cover maps for 2013 and 2023 and the patch-generating land use simulation (PLUS)-based projection for 2033; source: own study

Over the same period, dryland agriculture expanded by 42.37%, from 6,620 ha to 9,422 ha, and settlement area increased by 122.24%, from 1,785 ha to 3,964 ha. Shrubland decreased slightly by 2.37%, whereas rice fields increased by 5.84%, likely reflecting irrigation expansion. These shifts are consistent with broader tropical patterns of agricultural and urban encroachment and point to intensifying anthropogenic pressure within the watershed (Simbangala *et al.*, 2015; Abba *et al.*, 2021; Liu *et al.*, 2022). The patch-generating land use simulation (PLUS) model performed well, with a kappa coefficient of 0.89 and a figure of merit (FOM) of 0.68, indicating high spatial predictive accuracy consistent with Liang *et al.* (2021) and Liu *et al.* (2025).

The decline in forest cover is consistent with findings by Hu *et al.* (2023) and Pan, Wu and Liu (2015), who identified population growth and agricultural intensification as major drivers of deforestation and hydrological alteration. In Bolango, this conversion is especially important because replacing forest with agriculture or settlement increases surface runoff, weakens infiltration, and raises the probability of accelerated soil loss. As Widjonarko and Maryono (2022) noted, such transitions often increase sediment yield and reduce watershed resilience, underscoring the need for preventive land-management interventions rather than reactive rehabilitation alone.

Beyond the aggregate area changes, the spatial pattern of conversion shows that forest loss and agricultural expansion are concentrated where accessibility and land-allocation policies intersect, particularly along road-influenced corridors and designated cultivation zones, whereas remaining forest patches persist mainly in steep and less accessible terrain. This pattern supports the coupled socio-ecological framing adopted here: spatial planning instruments and their strong enforcement conditions where land conversion occurs and, consequently, where erosion risk intensifies.

EROSION RATE (RUSLE)

Rainfall erosivity (R) derived from climate hazards center infrared infrared precipitation with station data (CHIRPS) reflects annual precipitation between 340 and 3,500 mm across the

study area. Calculated R values ranged from 613.18 to 921.56 $\text{MJ mm ha}^{-1} \cdot \text{h}^{-1} \cdot \text{y}^{-1}$ in 2013 and from 582.63 to 760.80 in 2023–2033, indicating a moderate decline toward the projection period. Spatially, the highest erosivity values ($>900 \text{ MJ mm} \cdot \text{ha}^{-1} \cdot \text{h}^{-1} \cdot \text{y}^{-1}$) were concentrated in the upper Bolango sub-watershed, where orographic rainfall is strongest (Fig. 4). This pattern is consistent with studies reporting temporal variation in tropical rainfall aggressiveness and illustrates how climatic forcing continues to shape erosion risk even when land-cover change is the dominant local driver (Rakib, 2019; Ahmad *et al.*, 2022).

The spatial distribution of soil erodibility showed K values between 0.0264 and 0.0394, with the highest values (>0.035) concentrated in centrally located loamy soils. This range indicates that intrinsic soil properties contribute meaningfully to spatial differences in erosion susceptibility. The pattern agrees with broader evidence that finer-textured or structurally sensitive soils may be more erodible because of lower infiltration capacity and greater detachment potential (Zhang, Liu and Li, 2022; Liu, Rong and Wei, 2023). The use of OpenLandMap-derived soil information also improves the spatial precision of the K -factor map, consistent with the recommendations of Bouguerra *et al.* (2017) and recent data-driven soil mapping studies (Othmani *et al.*, 2023).

The LS factor derived from the 30 m Shuttle Radar Topography Mission digital elevation model (SRTM DEM) ranged from 0.1 to 276.82. The largest values occurred on slopes $>40\%$ in the upstream Bolango sector, confirming topography as a dominant control on erosion intensity (Fig. 4). This result aligns with Panagos, Borrelli and Meusburger (2015) and Kramm and Hoffmeister (2019), who showed that DEM resolutions of 30 m or finer are well suited for slope-based erosion analysis. It also agrees with Nikolova, Mitova and Dimitrov (2022) and Pereira *et al.* (2019), who emphasised the combined influence of slope steepness and slope length on erosion in complex terrain.

The C factor values derived from land-cover maps and the coefficients of Asdak (1995) and Arsyad (2010) further explain the spatial structure of erosion risk. Forested areas retained very low C values (0.001–0.003), whereas shrubland, agricultural land,

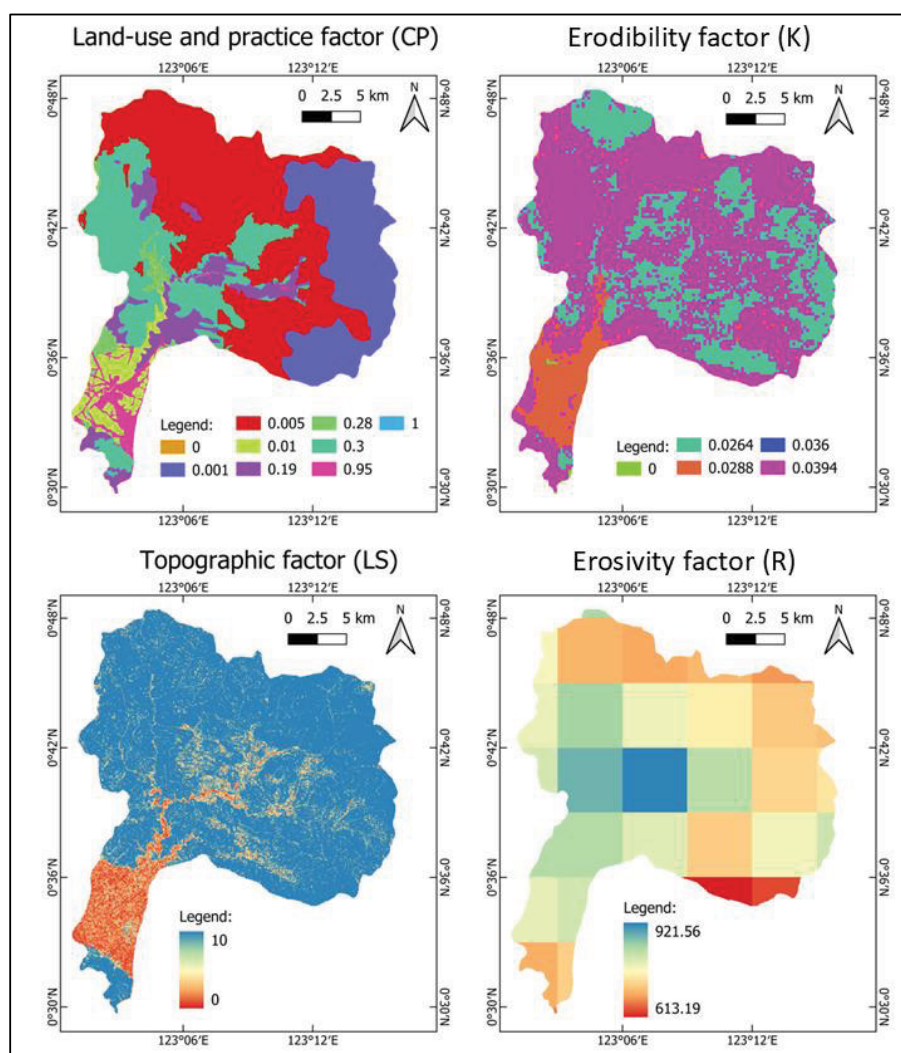


Fig. 4. Spatial distribution of the main revised universal soil loss equation (RUSLE) factors (R , K , LS , and CP) across the Bolango Watershed; source: own study

and settlements showed progressively higher values (0.1, 0.3–0.5, and 0.8–0.9, respectively). Thus, conversion from forest to cultivated or built-up land substantially increases exposure to erosion. This pattern is consistent with global evidence linking vegetation loss and declining normalised difference vegetation index (NDVI) to higher erosion potential (Kidane *et al.*, 2019).

The P factor, derived following David (1988) and Benavidez *et al.* (2018), ranged from 0.005 in protected forest to 1.0 in open or urbanised land, highlighting strong spatial variation in conservation effectiveness across slope and land-use settings. Lower P values correspond to areas where conservation practices or protective land cover limit erosion, whereas higher values indicate minimal protection under intensive human use. This pattern reinforces the importance of contour farming, vegetative buffering, and slope-based zoning as practical erosion-control measures in the Bolango Watershed (Iticha and Takele, 2018; Sud *et al.*, 2024).

Mean soil erosion estimated with RUSLE was $59.08 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2013 and $49.22 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2023. The erosion-severity maps (Fig. 5) show persistent hotspots in the upper watershed and in dryland agricultural zones, mirroring patterns reported for other tropical catchments (Panagos, Borrelli and Meusburger, 2015). Critically eroded land ($>480 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$)

declined from 7,496.14 ha in 2013 to 6,250.23 ha in 2023, while overall sediment yield fell by 16.7%. Although these trends indicate modest improvement, they also show that land degradation pressure remains substantial rather than resolved (Borrelli *et al.*, 2020; Payuyu, 2023).

The reduction in mean erosion from 2013 to 2023 is best explained by the combined effect of declining R and the contraction of the most severely eroded class area, which together offset the erosive influence of continued land conversion. In other words, the model suggests that climatic forcing and the spatial redistribution of high-risk cells temporarily outweighed the negative effects of land-use pressure at the watershed scale, even though local hotspots persisted.

AGENT-BASED MODELLING–REVISED UNIVERSAL SOIL LOSS EQUATION, INTEGRATION FOR EROSION RATE SIMULATION

Erosion simulations for 2033 revealed clear divergence among management scenarios. Scenario 1 (business-as-usual) produced total erosion of $2.63 \text{ mln Mg}\cdot\text{y}^{-1}$, representing the continuation of current land-use practices. Within the ABM, farmers, government officials, and investors interacted over a 10–30-year horizon, and their decisions were linked to erosion outcomes

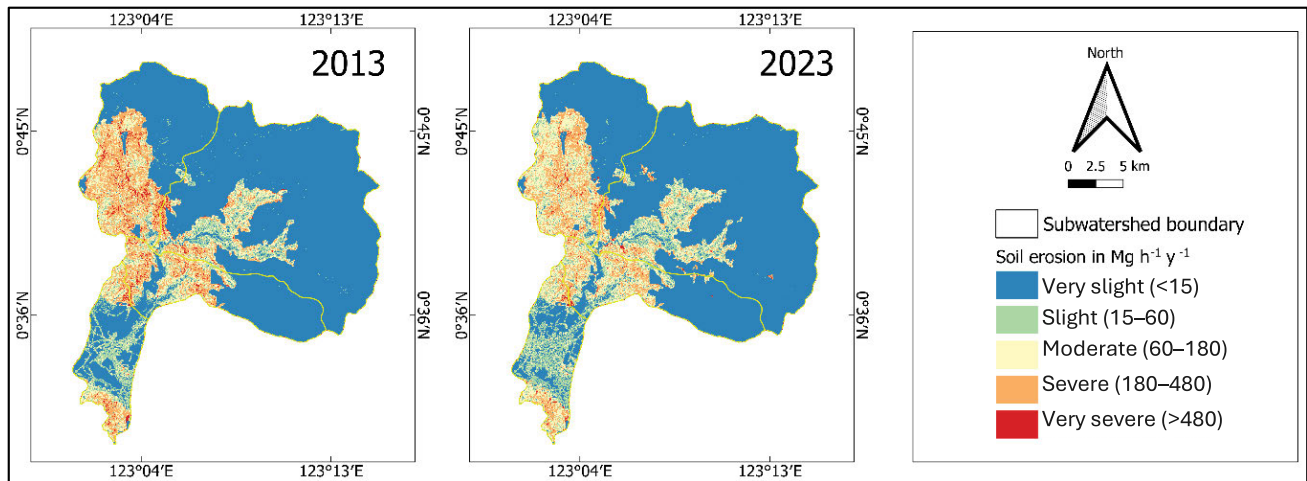


Fig. 5. Erosion-severity classes mapped for 2013 and 2023 from the revised universal soil loss equation (RUSLE) model; source: own study

through adaptive feedbacks. Farmer behaviour was especially influential because decisions on land conversion and conservation directly altered the C and P values used in RUSLE. Interview evidence indicated that contour-aligned planting is the most common conservation practice currently recognised by farmers; this insight informed scenario 2, in which P values were set to 0.6 on 9–12° slopes, 0.7 on 13–16° slopes, 0.8 on 17–20° slopes, 0.9 on 21–25° slopes, and 0.95 on slopes >25° (David, 1988). Under this scenario, total erosion declined to 1.92 mln $\text{Mg}\cdot\text{y}^{-1}$, a 27% reduction (Tab. 1). These results confirm that spatially targeted contour-based conservation can materially reduce erosion risk in

tropical landscapes, consistent with Madhukar *et al.* (2023) and Woldemariam *et al.* (2018).

Within the coupled ABM–RUSLE framework, agents affect the scenarios through two direct pathways: (i) land-use transitions, such as forest-to-agriculture or agriculture-to-settlement conversion, which update the C factor; and (ii) adoption or enforcement of conservation measures, such as contour farming or slope protection, which update the P factor. At each time step, the revised C and P layers are passed to RUSLE to compute annual average soil loss (A), and the resulting erosion levels feed back into agent attributes, including awareness and perceived

Table 1. Distribution of watershed area by erosion-severity class in 2013, 2023, and projected 2033

Sub watershed	Area (ha)					
	class 1 (very low)	class 2 (low)	class 3 (moderate)	class 4 (high)	class 5 (very high)	total
2013						
Mongiilo	22,089.51	1,303.92	1,634.58	973.35	177.39	26,178.75
Bolango Hilir	2,574.09	1,330.11	1,013.31	640.89	124.29	5,682.69
Langge	4,621.95	622.62	1,188.63	1,286.82	187.92	7,907.94
Dulamayo	6,758.46	514.89	1,988.19	3,013.83	426.69	12,702.06
Total	36,044.01	3,771.54	5,824.71	5,914.89	916.29	52,471.44
2023						
Mongiilo	21,756.51	1,504.08	1,891.98	900.72	125.46	26,178.75
Bolango Hilir	2,006.73	1,849.32	1,191.60	545.49	89.55	5,682.69
Langge	4,516.38	830.43	1,501.56	973.71	85.86	7,907.94
Dulamayo	6,630.57	803.97	2,828.34	2,276.37	162.81	12,702.06
Total	34,910.19	4,987.80	7,413.48	4,696.29	463.68	52,471.44
2033						
Mongiilo	21,726.63	1,490.76	1,920.87	919.98	120.51	26,178.75
Bolango Hilir	1,567.08	2,146.14	1,294.38	569.34	105.75	5,682.69
Langge	4,419.00	867.87	1,579.86	950.67	90.54	7,907.94
Dulamayo	6,611.58	801.27	2,859.03	2,269.17	161.01	12,702.06
Total	34,324.29	5,306.04	7,654.14	4,709.16	477.81	52,471.44

Source: own study.

risk, thereby shaping subsequent decisions. This structure makes the social-to-biophysical coupling explicit and transparent.

Scenario 3 represented the implementation of protected-area regulations on slopes >25% under Indonesian law (Undang-Undang, 2007) on spatial planning. In this scenario, government action is decisive because regulatory enforcement determines whether steep land remains protected and whether the corresponding *CP* value is reduced to 0.005. Scenario 3 produced the greatest ecological stability, lowering land-use fluctuation by 20% relative to business-as-usual. It also achieved the strongest erosion reduction, decreasing total soil loss to 1.28 mln $\text{Mg}\cdot\text{y}^{-1}$, or 51% below the business-as-usual scenario. Model validation yielded $R^2 = 0.83$ and $\text{RMSE} = 12.7 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$, consistent with the validation standards of Thapa (2020) and Kwang *et al.* (2022). These results reinforce the value of ABM for integrating socio-economic behaviour into environmental forecasting (Findlater, Satterfield and Kandlikar, 2019; Tanaka, Hanley and Kühfuss, 2022) – Figure 6.

Pearson correlation analysis showed that annual average soil loss (*A*) was positively correlated with *R* ($r = 0.64$), *LS* ($r = 0.78$), and *C* ($r = 0.82$), and was negatively correlated with *P* ($r = -0.65$). Sensitivity ranking followed the order $C > LS > R > P > K$, indicating that vegetation management exerts the strongest control on erosion variability. This hierarchy is consistent with Zhu *et al.* (2019) and Ugeke *et al.* (2022), who also identified vegetation cover as the dominant driver in soil-erosion modelling. The strong positive relationship between *LS* and *C* ($r = 0.79$) further highlights the compounding effect of steep slopes and vegetation loss on total soil loss (Erdoğan, Esbah and Berberoğlu, 2016).

The integrated ABM–RUSLE model showed strong predictive skill, with a mean absolute percentage error (*MAPE*) of 8.56% and an R^2 of 0.83, indicating good agreement between predicted and observed erosion rates. Visual comparison of erosion maps also showed close spatial correspondence between observed and simulated hotspots, similar to validation approaches reported by Gebre, Belete and Belayneh (2024). The joint use of statistical and spatial validation strengthens model robustness (Chen *et al.*, 2019; Raj *et al.*, 2024). Relative to earlier studies in comparable settings (Payuyu, 2022; Borrelli *et al.*, 2020), the coupled framework performs better for dynamic erosion assessment because it integrates both biophysical controls and human decision-making.

Taken together, these results demonstrate the value of combining empirical and behavioural modelling for erosion prediction under changing land-use conditions. The simulations quantify the benefits of specific conservation interventions and provide a stronger basis for prioritising practical watershed-management strategies in tropical environments.

SPATIAL MANAGEMENT RECOMMENDATIONS

Spatial management recommendations for the Bolango Watershed were derived from the coupled ABM–RUSLE simulations under three land-use management scenarios: business-as-usual (scenario 1), contour farming implementation (scenario 2), and protected-area regulation (scenario 3). Considering the three scenarios jointly makes it possible to translate modelled erosion outcomes into spatial priorities for sustainable watershed management (Fig. 7).

From an institutional perspective, the scenarios represent different levels of coordination and enforcement involving local governments, provincial planning and environmental agencies, watershed authorities, and farmer organisations. Scenario 1 reflects continuation of existing practice with limited additional enforcement. Scenario 2 assumes stronger extension support and incentives that enable farmer groups to adopt contour-based practices at scale. Scenario 3 assumes strict implementation of protected-area zoning on steep slopes, requiring coordinated permitting, monitoring, and compliance. This clarification makes explicit that agency roles are integral to the management rules tested by the model.

• Scenario 1: Business-as-usual

Scenario 1 (business-as-usual – BAU) was conducted using the 2033 land cover and without the implementation of additional conservation measures. This scenario represents natural land-use change patterns following historical trends. Interventions in the form of conservation policies, land rehabilitation, or the application of soil and water conservation techniques were not implemented in this scenario. Therefore, this scenario served as the baseline for evaluating the conservation scenarios.

• Scenario 2: Contour-based farming system

Under scenario 2, contour-based cultivation reduced soil loss across many of the high-risk agricultural areas. The analysis identified approximately 8,050 ha of class 3 and 1,662 ha of class 4 land that require immediate intervention, especially in the

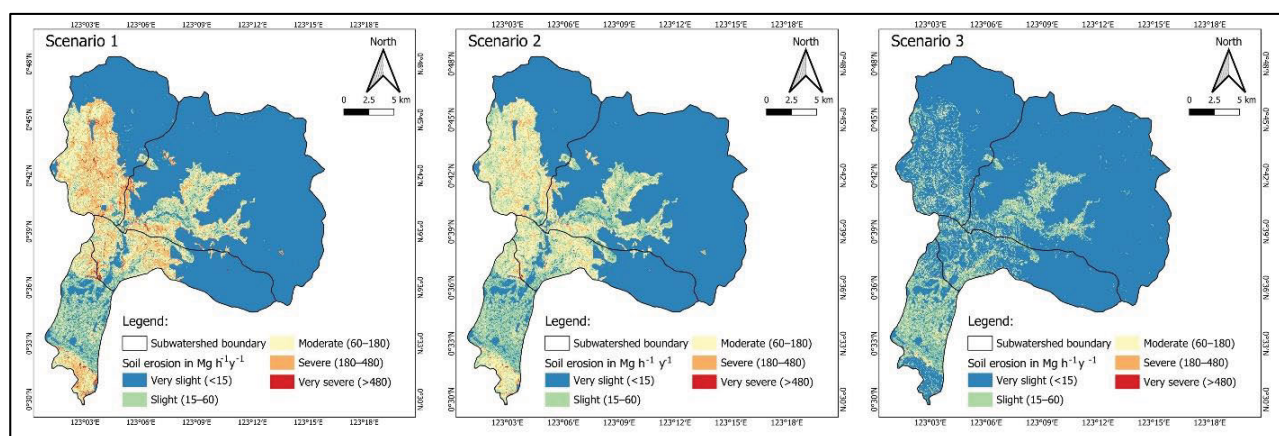


Fig. 6. Comparison of the 2033 scenarios, showing how business-as-usual, contour farming and protected steep-slope management alter the magnitude and spatial distribution of predicted erosion; scenarios 1–3 as in pp. 235–236; source: own study

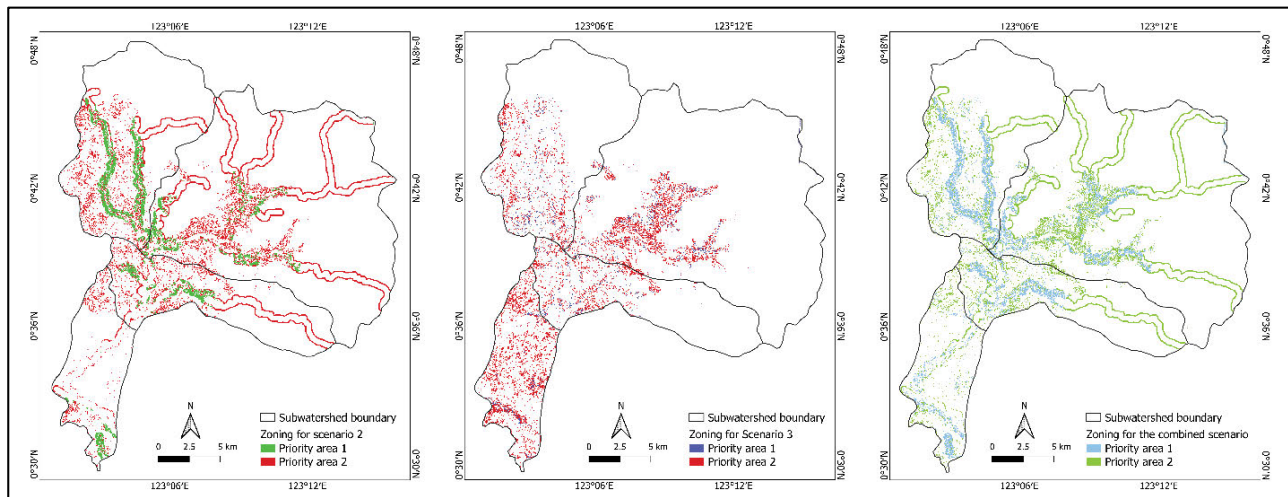


Fig. 7. Priority-zone map for land conservation in the Bolango Watershed derived from the scenario analysis, highlighting where protection, restoration, and sustainable farming practices are most critical for reducing erosion; scenarios 2 and 3 as in pp. 235–236; source: own study

Dulamayo and Mongiilo sub-watersheds. A smaller but still critical area of about 190 ha remained in class 5, mainly along steep slopes and river corridors. These patterns suggest two priority zones. Priority zone 1 (ZP-1), comprising slopes $>25\%$ and areas within 300 m of major rivers, requires structural terracing, vegetated contour strips, micro check dams, and rapid establishment of stabilising cover such as leguminous ground cover and fast-growing trees. Priority zone 2 (ZP-2), covering moderate slopes (9–25%) and adjacent transition areas, should emphasise strip cropping, mulching, organic soil improvement, and incentive programs that enable broad adoption by farmer groups.

• Scenario 3: Enforcement of protected-area regulation

Scenario 3, which enforces protected-area zoning on slopes $>25\%$ in accordance with Indonesian law (Undang-Undang, 2007), produced the most substantial erosion reduction. More than 83.8% of the watershed shifted into the very low erosion-risk class (class 1), whereas the high and very high erosion classes were reduced to less than 0.5% of the total area. In this scenario, priority zone 1 includes steep slopes and critical upper catchments in Mongiilo, Dulamayo, Bolango Hilir, and Langge. Recommended actions are permanent green protection through reforestation with native species, cover crops, and strict prohibition of land clearing, supplemented by bioengineering measures such as bench terraces and vegetated check structures along runoff pathways. Priority zone 2 comprises surrounding buffer areas on 13–25% slopes, where agroforestry and micro-contour drainage should be promoted to maintain low erosion levels while supporting regulated land use within the regional spatial plan framework.

• Cross-scenario spatial priorities

Across scenarios, all class 4–5 cells and class 3 areas on slopes $\geq 15\%$ or within 300 m of rivers should be treated as critical conservation zones. These locations require permanent vegetative cover, riparian greenbelts 30–50 m wide, and targeted soil-bioengineering measures. By contrast, class 2 areas on moderate slopes (9–25%) function as preventive management zones, where sustainable cultivation practices and regular monitoring can prevent escalation to higher erosion classes.

• Implementation and monitoring targets

For implementation, the model suggests two near-term performance targets: reducing erosion in ZP-1 by at least 80% within 3–5 years through conversion of class 3–5 areas to class

1–2 and maintaining at least 85% of ZP-2 within stable erosion classes through continuous practice audits and farmer-compliance checks. Updating the spatial management maps every two years would allow land-cover shifts and conservation effectiveness to be tracked systematically.

These recommendations provide an evidence-based framework for integrating biophysical risk and socio-behavioural drivers in watershed management. Long-term resilience in the Bolango Watershed will depend not only on technical conservation measures, but also on sustained farmer engagement and consistent policy enforcement across institutions.

Between 2013 and 2033, the Bolango Watershed experienced a marked intensification of human pressure, reflected in a 14.25% reduction in forest area and major expansions of dryland agriculture (42.37%) and settlements (122.24%). In a watershed with steep topography, such shifts are especially consequential because they increase runoff generation and sediment mobilisation. As vegetation cover declines, the *C* factor increases, and where conservation practices weaken, the *P* factor also rises, together elevating erosion risk across cultivated and peri-urban zones.

Despite these pressures, mean soil loss decreased from $59.08 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2013 to $49.22 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2023 and then stabilised at $50.17 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2033. This modest net improvement is attributable mainly to lower modelled rainfall erosivity and some conservation uptake, rather than to a fundamental reduction in development pressure. Hotspots remain concentrated in upper-catchment areas with high *LS* values and limited vegetative cover, confirming that slope and land cover remain the dominant controls on erosion in the basin.

The scenario analysis demonstrates that targeted conservation measures can materially change erosion trajectories. Under business-as-usual, total soil loss reaches $2.63 \text{ mln Mg}\cdot\text{y}^{-1}$, whereas contour farming lowers this by 27% and protection of slopes $>25\%$ lowers it by 51%. The ABM further shows that these gains depend on behaviour: policy enforcement, institutional coordination, and environmental awareness directly affect conservation adoption and land-use stability. The 20% reduction in land-use fluctuation under scenario 3 indicates that strict zoning not only reduces erosion, but also strengthens socio-ecological resilience.

Sensitivity analysis identified the *C* factor as the most influential variable, followed by *LS* and *R*, indicating that vegetation management and slope regulation are the most effective leverage points for erosion control. The immediate policy implication is clear: steep-slope protection, contour farming, and restoration of degraded upper catchments should be prioritised within watershed planning and implementation. These priorities are consistent with Indonesian spatial planning regulations under Indonesian law (Undang-Undang, 2007) and provide a practical basis for targeted intervention.

Overall, the integrated ABM–RUSLE framework offers a dynamic and policy-relevant way to link human behaviour with biophysical erosion processes. By capturing feedback between land-use change and erosion, the model provides evidence that is directly usable for planning, scenario evaluation, and adaptive watershed management. Future work should extend the framework through field-based sediment calibration and tighter coupling with hydrological or ecosystem-service models such as strengths, weaknesses, opportunities, and threats (SWAT) or integrated valuation of ecosystem services and tradeoffs (InVEST) to assess downstream sediment delivery and broader ecological impacts.

CONCLUSIONS

This study demonstrates that soil erosion in the Bolango Watershed emerges from the combined effects of land-use change, topography, and human decision-making, and that these interactions can be represented effectively through an integrated agent-based modelling–revised universal soil loss equation (ABM–RUSLE) framework. Over 2013–2033, forest loss of 14.25%, expansion of agriculture by 42.37%, and settlement growth of 122.24% increased erosion susceptibility, particularly in steep upper catchments where vegetation cover is declining. The RUSLE estimates yielded mean erosion rates of $59.08 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2013, $49.22 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2023, and $50.17 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ in 2033, indicating a modest overall decrease driven mainly by reduced rainfall erosivity and localised conservation effects. Nevertheless, the persistence of hotspots above $480 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$ confirms that critical slope areas remain highly vulnerable.

Scenario analysis highlights the practical value of conservation measures: contour farming reduced total soil loss by 27%, whereas protection of slopes $>25\%$ reduced it by 51%. The ABM component further showed that governance quality and community participation are decisive for sustaining those gains because they influence whether conservation practices are actually adopted and maintained. Model validation ($R^2 = 0.83$; $\text{RMSE} = 12.7 \text{ Mg}\cdot\text{ha}^{-1}\cdot\text{y}^{-1}$) confirmed strong predictive performance, while sensitivity analysis identified the cover-management factor (*C*) as the dominant source of erosion variability.

This research advances soil-erosion assessment by embedding socio-behavioural dynamics within an empirical biophysical model. The result is a replicable framework for adaptive watershed management under continuing land-use pressure. Future research should strengthen calibration with field-based sediment-yield measurements and couple the present framework with hydrological models such as SWAT or ecosystem-service models such as InVEST to evaluate downstream sediment transport and broader environmental effects.

In practical terms, the findings support two immediate planning priorities: first, protect and restore steep upper-catchment areas where high *LS* values amplify erosion response; second, scale up farm-level conservation on moderate slopes to stabilise cultivated landscapes. These priorities provide a concrete basis for integrating erosion control into watershed planning and spatial policy in the Bolango Watershed.

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CONFLICT OF INTERESTS

All authors declare that they have no conflict of interests.

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